

Exercise 2

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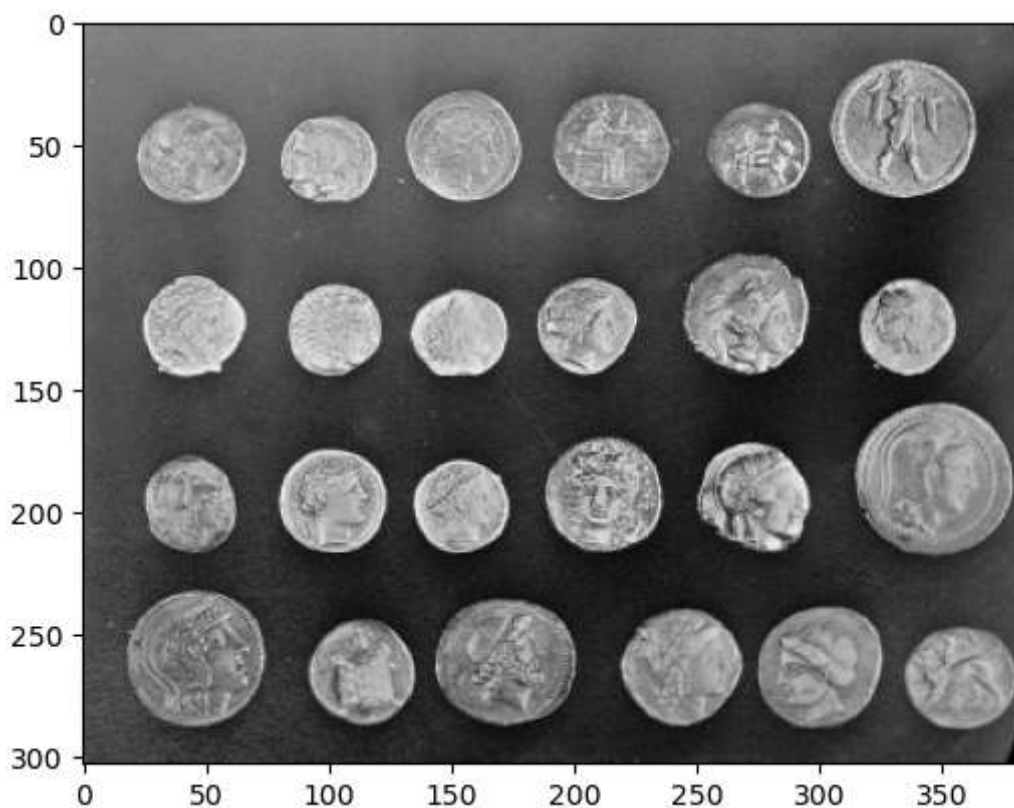
```
In [ ]: !mamba install scikit-image
```

```
In [5]: import matplotlib.pyplot as plt
import numpy as np
import skimage as ski
```

a) Load the original image (top row of Fig. 1) and take its logarithm. Make sure that your code never attempts to take the logarithm of zero. (2P)

```
In [21]: image = ski.img_as_float(ski.data.coins()) + 0.001
log_image = np.log(image)
plt.imshow(image, cmap = 'gray')
```

```
Out[21]: <matplotlib.image.AxesImage at 0xb3649a0>
```

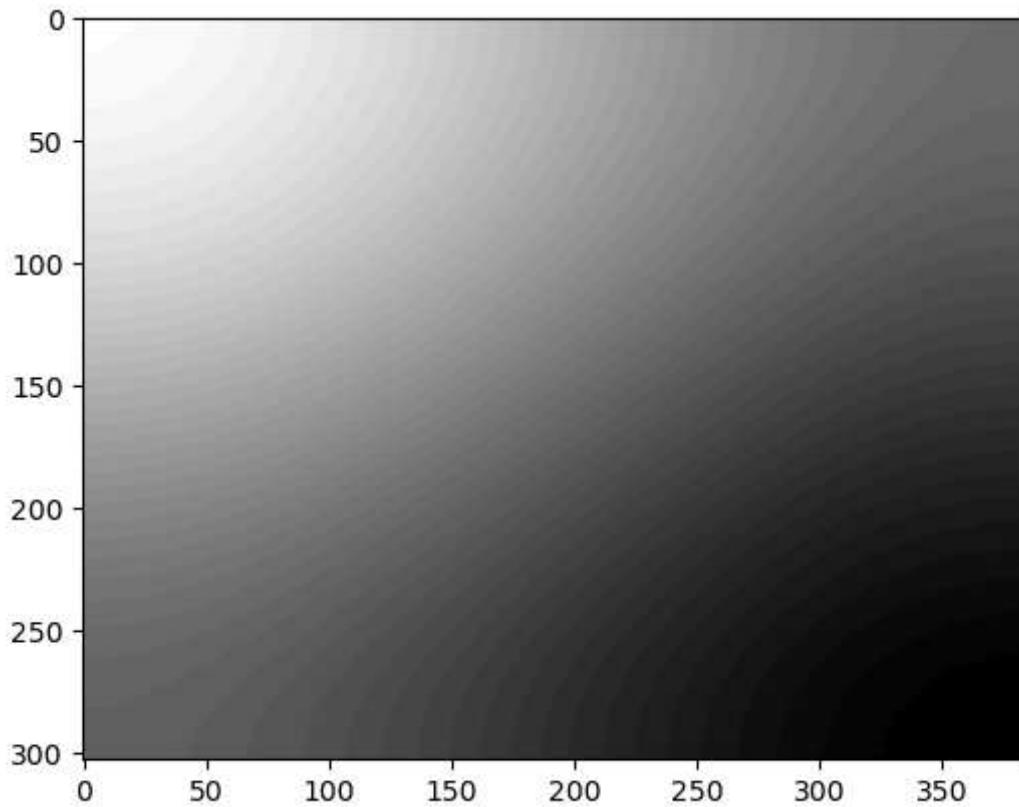


b) Compute the logarithms of at least two different smoothed versions of the image, selecting suitable scales. (2P)

```
In [18]: from scipy.ndimage import gaussian_filter

smooth_image_1 = gaussian_filter(image, 10)
smooth_image_2 = gaussian_filter(image, 200)
smooth_log_1 = np.log(smooth_image_1)
smooth_log_2 = np.log(smooth_image_2)
```

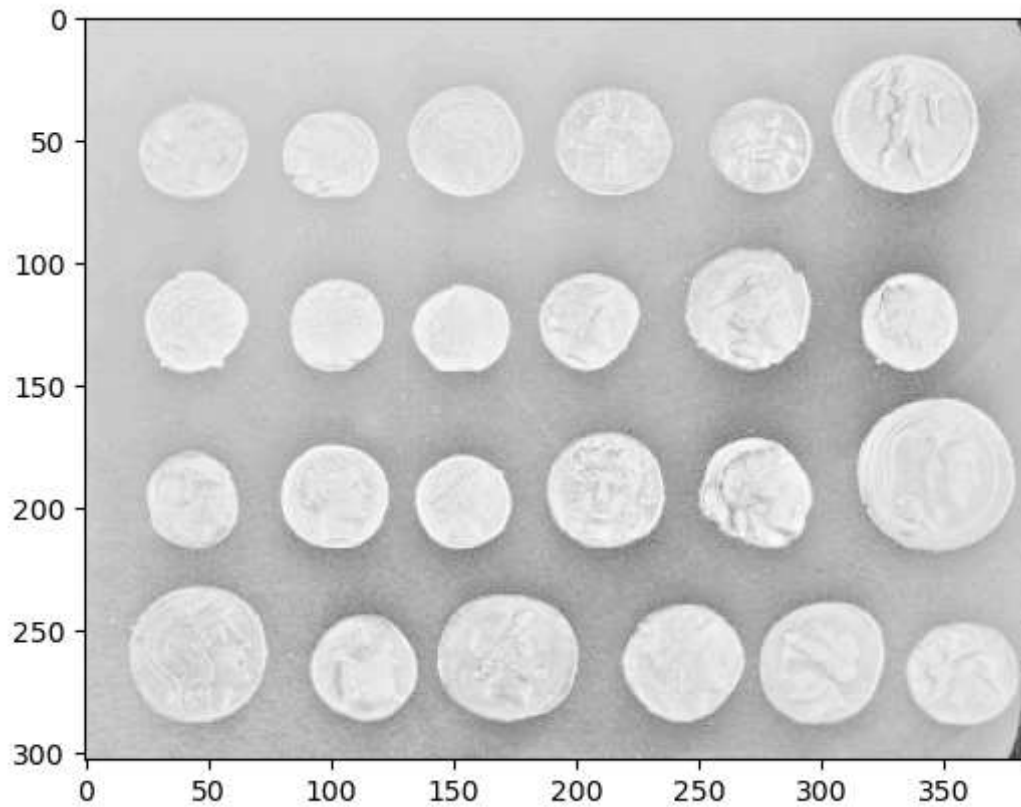
```
Out[18]: <matplotlib.image.AxesImage at 0xb082320>
```



c) Evaluate Eq. (1) and plot the resulting retinex image F after converting it back to the expected numerical range of an image via min-max-scaling. At this point, your result might look similar to the one in the middle row of Fig. 1, i.e., the contrast might still be worse than in the original. (2P)

```
In [31]: processed_image = 3*log_image-smooth_log_1-2*smooth_log_2
min_val = np.min(processed_image)
max_val = np.max(processed_image)
retinex_image = (processed_image-min_val)/(max_val-min_val)
plt.imshow(retinex_image, cmap = 'gray')
```

```
Out[31]: <matplotlib.image.AxesImage at 0xc8eeaa0>
```



d) Perform an alternative re-scaling of the retinex image by clipping some small percentage of the most extreme values, instead of just taking the minimum and maximum. Your result should now look similar to the bottom row of Fig. 1. (2P)

```
In [40]: alpha = 0.01
retinex_image_2 = (processed_image-alpha*min_val)/(max_val-alpha*min_val)
retinex_image_2 = np.clip(retinex_image_2, 0, 1)
plt.imshow(retinex_image_2, cmap = 'gray')
```

```
Out[40]: <matplotlib.image.AxesImage at 0xdcd9848>
```

